

Hands-On Lab: Sprint 1 Planning & Full Schema Design

Project Name: Cardiac Patient Monitoring System (Healthcare Management API)

Target Milestone: Phase 3 - Sprint 1

1. Sprint 1 Goal & Backlog Board Setup

- **Sprint 1 Goal Statement:**

"Establish a normalized, secure database schema with Entity Framework Core and build paginated retrieval endpoints for patients alongside a transaction-backed appointment scheduling pipeline."

- **Backlog Board Columns (Notion/Trello):**

- **Backlog:** Sized tasks waiting for prioritization.
 - **To Do:** Confirmed tasks committed to Sprint 1.
 - **In Progress:** Active development tasks.
 - **Peer Review / Pull Request:** Code committed to GitHub awaiting review.
 - **Done:** Completed tasks that meet the Definition of Done (DoD) [4, 8].
-

2. Complete Entity List (Full Professional Baseline)

To establish a production-ready healthcare baseline, the database schema incorporates the following entities:

1. **IdentityUser (AspNetUsers):** Manages user credentials, password hashing, and system login identities.
2. **IdentityRole (AspNetRoles):** Manages system roles (Admin, Doctor, Patient) to enforce Role-Based Access Control (RBAC).
3. **Patient:** Houses patient demographic profiles, contact information, and relates strictly to a unique system user identity.
4. **VitalSign:** Tracks patient telemetry readings (Heart Rate, Oxygen Saturation, Systolic and Diastolic Blood Pressure) taken over time.

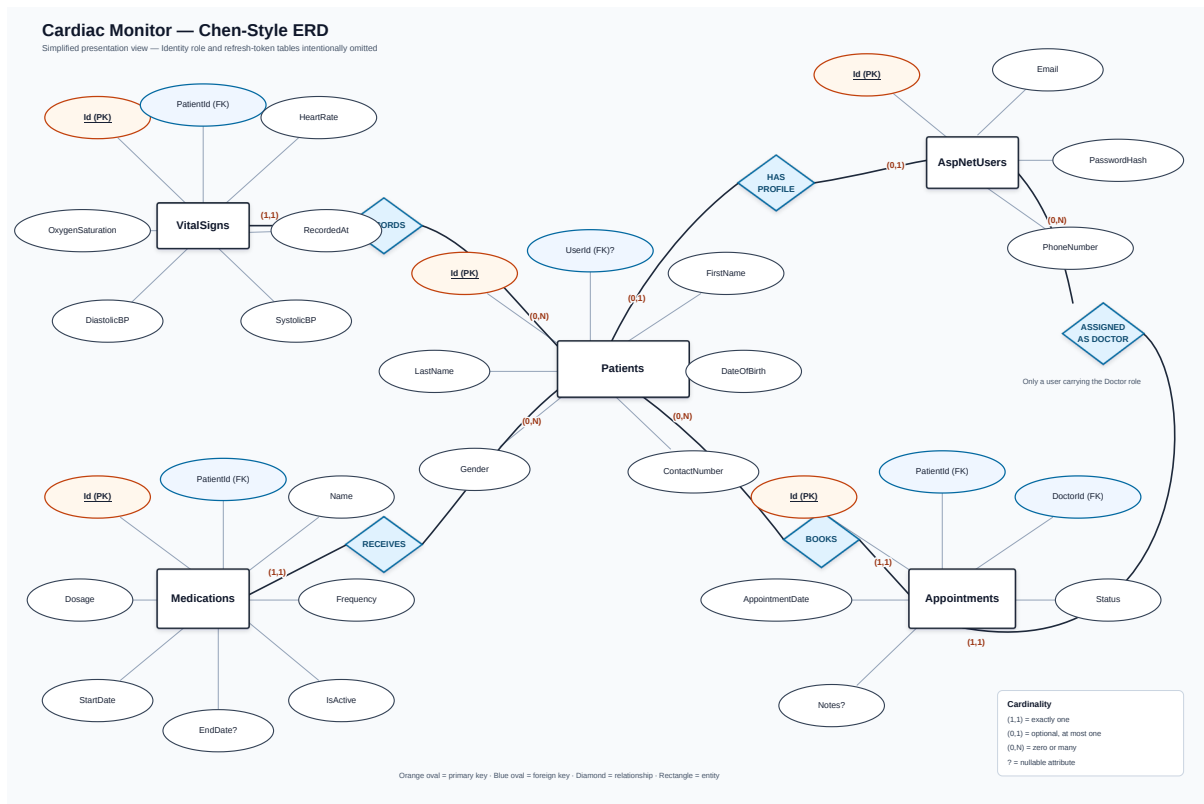
5. **Medication:** Manages active and historical prescription lists, including dosage, frequency, and active timelines.
 6. **Appointment:** Schedules clinical consultations between a patient and a doctor (Identity User).
 7. **RefreshToken:** Securely persists stateless session tokens to allow secure access-token rotation without re-authentication.
-

3. Normalized Schema Design (3NF Compliance)

The schema adheres strictly to Third Normal Form (3NF) to eliminate data redundancy and prevent anomalies [4]:

- **First Normal Form (1NF):** Every column stores atomic, single-valued attributes (e.g., separating FirstName and LastName, ensuring no comma-separated text lists) [4].
 - **Second Normal Form (2NF):** All tables have a clearly defined primary key (Id), and all non-key attributes depend entirely on the primary key, eliminating partial dependencies [4].
 - **Third Normal Form (3NF):** Non-key attributes depend only on the primary key, with no transitive dependencies [4]. For example:
 - Doctor details are referenced solely via DoctorId toAspNetUsers, preventing the duplication of physician profiles in the Appointments table [4].
 - Patient demographics remain isolated in the Patients table, avoiding duplication across clinical tables [4].
-

4. Finalized Entity Relationship Diagram (ERD)



• Relationship Cardinals & Rules:

- Patients to VitalSigns, Medications, and Appointments are configured with **Cascade Delete** rules to automatically clean up medical records if a patient profile is deleted [5].
- Appointments to AspNetUsers (Doctors) is configured with **Restrict Delete** rules to prevent database exceptions and circular cascade paths [5].

5. Sprint 1 Sized Backlog Tasks

Monolithic requirements are broken down into granular, estimable backlog items of half-a-day (0.5 day) to a full day (1.0 day) to facilitate productive daily stand-up reporting [4]:

Status	Estimated Effort	Task Description	Task ID
To Do	Day 0.5	Implement Patients retrieval endpoint with pagination (Skip & Take LINQ operations) [6]	TSK-101
To Do	Day 0.5	Add optional search and filtering query parameters (Gender, Name)	TSK-102

		.to the Patients list [6]	
To Do	Day 0.5	Implement dynamic sorting (by LastName or DateOfBirth with .ASC/DESC options) [7]	TSK-103
To Do	Day 0.5	Implement business logic check for overlapping doctor appointments [7]	TSK-104
To Do	Day 0.5	Wrap multi-step appointment creation in an EF Core Database Transaction (.BeginTransactionAsync) [7]	TSK-105
To Do	Day 1.0	Write integration tests using WebApplicationFactory for paginated and transaction-backed endpoints [6, 7]	TSK-106